Course Overview

CSC 115: Fundamentals of Programming II

Anthony Estey



Territorial Acknowledgement

We acknowledge and respect the Lək̓ʷəŋən (Songhees and Xʷsepsəm/Esquimalt) Peoples on whose territory the university stands, and the Lək̓ʷəŋən and W̱SÁNEĆ Peoples whose historical relationships with the land continue to this day.

Who am I?

- Anthony Estey – Instructor of sections A01 – A09
  - From a small town, Sechelt, here in BC, Canada
  - I completed my Ph.D. at the University of Victoria in 2016
  - Was an instructor at UBC from September 2016 – August 2019

- Since returning to UVic, I have taught CSC 106, CSC 110, CSC 111, CSC 115, CSC 130, and CSC 225
  - I first taught the Java version of CSC 115 in the Summer 2020 term, and taught it almost every term from until the January 2026 term
  - I developed the new C++ version, first offered in the Fall 2026 term

Who are you?

- What computer science course did you take before entering CSC 115? (show of hands)

A. CSC 110 at UVic
B. CSC 111 at UVic
C. A programming course during high-school
D. A programming course at a different university/college
E. Other

Who are you?

- How long ago did you take your prerequisite course?

A. Last term
B. Within the last 12 months
C. Between 12 and 24 months
D. More than 2 years ago

Who are you?

- Have you ever programmed in C++ before?

A. Yes, I have quite a bit of experience programming in C++
B. Yes, but only a little bit
C. Nope

Goals of this course

- Build on the fundamentals gained in your previous course
  - You already have the tools: **variables, loops, arrays, and functions**.
  - Because you know how to use them, our focus shifts this term:
    - **From:** *How do we implement a correct solution to this problem?*
    - **To:** *What is the **best** (of possibly many) solutions to this problem?*
- In the real world, a program that is merely "correct" isn't good enough if it drains a phone's battery, is a mess for a others to read, or has to be completely rewritten in a week
  - This term, we will learn to think like a software architect by focusing on three main pillars.

Goals of this course

1. **Readability**
   - **The Reality:** Code is read far more often than it is written
   - **Our Goal:** Continue with the function design, documentation, and style guidelines from your previous course
   - **The Takeaway:** Write clean, clear code that other developers (and our future selves) can easily use and understand at a glance
2. **Reusability**
   - **The Reality:** Programs are actively used, patched, and extended–or they are dead
   - **Our Goal:** Learn to use and write code that can be easily reused in future projects
   - **The Takeaway:** Stop reinventing the wheel every time we start a new project
3. **Efficiency**
   - **The Reality:** Software is dependent on the hardware it runs on
   - **Our Goal:** Learn to analyze the runtime speed and memory usage of our code
   - **The Takeaway:** Build high-performance programs that can handle massive amounts of data without slowing down

Course Format

- Grade Breakdown:

Component	Weight
Participation	1%
Pre-Lecture Quizzes	5%
Labs	9%
Assignments	20%
Exams	65%
TOTAL	**100%**

Course Format

- Participation (1%)
  - For things that you *should* do anyways (like a practice exam)

- Lectures
  - Before lecture, you watch pre-lecture videos, and then complete a quiz to verify you understood the material (5%)
  - The assumption is that you have watched these videos and completed the quiz before lecture
  - Lecture time is spent digging deeper into these topics, often via code traces, code writing, or algorithm exploration activities

Course Format

- Weekly assessments
  - Most weeks there is a lab (9%) where you get a chance to complete activities similar to what we did together in lecture
  - Each week typically also has an assignment (20%) which is due by Friday night at midnight, which goes even deeper into each topic
  - Labs and assignment files are submitted through PrairieLearn
  - Labs will also have an oral component. To get credit for that, you will answer TA questions during the lab

Course Format

- Exams (65%):
  - There are three midterm exams and one final exam this term
  - Your lowest midterm exam is replaced by your final exam (if your final exam is higher than at least one midterm)
- All exams will be written in the CBTF (ECS 266):
  - You will complete your exam using PrairieTest, which has an interface similar to PrairieLearn
  - There are practice problems posted on PrairieLearn to get you used to the interface
  - There will also be an official practice exam (more info later)

Resources

- BrightSpace page: https://bright.uvic.ca
  - All lecture material will be posted on the BrightSpace course page
  - Pre-lecture video links, quizzes, slides, announcements, etc.
  - You should all be automatically added to the BrightSpace course page

- PrairieLearn:
  - You will submit lab and assignment code files to PrairieLearn
  - Practice problems can be found on PrairieLearn
  - You will be writing exams through PrairieTest (a PrairieLearn extension)

- No textbook required for this course